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Roll No.

PAPER ID—11182

B.Tech. EXAMINATION, 2024

(Fifth Semester)

ARTIFICIAL INTELLIGENCE AND DATA
SCIENCE

Machine Learning and its Applications

Time : 3 Hours

Maximum Marks : 70

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Unit. Q. No. 1 is compulsory. All questions carry equal marks.

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1. (a) What is the role of validation data in machine learning ?
- (b) List out any *five* real-time applications of Machine Learning.
- (c) How does Linear Regression differ from Logistic Regression in terms of their applications.
- (d) Define Binary Classification.
- (e) What is the F1-score.
- (f) What is Minkowski distance.
- (g) Differentiate between bagging, boosting and voting. $\frac{6}{2 \times 7 = 14}$

Unit I

2. (a) What are the key stages in the machine learning model life-cycle ? Explain with suitable diagram. 7

- (b) How do you determine when to stop tuning a machine learning model ? What are the consequences of over-tuning or under-tuning a model ? 7

3. (a) Discuss the major differences between statistical modeling and machine learning, focusing on their objectives, assumptions and approach to data analysis, providing examples to support your explanation. 7
- (b) Differentiate between Supervised, Unsupervised, and Reinforcement Learning with suitable examples for each. 7

Unit II

4. What is feature normalization and why is it important for machine learning models ? Explain at least seven methods of feature normalization with suitable mathematical description. 14

5. What is Principal Component Analysis (PCA) and how does it reduce the dimensionality of data ? Explain the steps involved in performing PCA on a dataset. 14

Unit III

6. Discuss the concepts of Entropy and Information Gain as applied in decision tree algorithms. Using a dataset aimed at predicting whether a customer is likely to 'Buys_computer' (Class Label), determine the most suitable attribute among age, income, student and credit_rating for the primary split according to the ID3 algorithm. Design the tree upto two levels atleast. 14

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Age	Income	Student	Credit rating	Buys_computer
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle aged	Low	Yes	Excellent	Yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle aged	Medium	No	Excellent	Yes
Middle aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

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P.T.O.

7. What is Support Vector Machine (SVM). Differentiate between hard SVM and Soft SVM along with suitable diagrammatical and mathematical representation. Enlist five kernel functions with corresponding mathematical equations and explain their role in SVM based classification problems. 14

Unit IV

8. (a) Illustrate K means clustering algorithm with an example. 7
- (b) Explain Singular Value Decomposition (SVD) and its steps for dimensionality reduction in machine learning. Illustrate its application with an example showing how SVD improves computational efficiency. 7

9. Consider the following confusion matrix for a binary classification problem, where 'Positive' refers to the positive class and 'Negative' refers to the negative class :

	Predicted Positive	Predicted Negative
Actual Positive	50	10
Actual Negative	5	35

Calculate the following performance metrics based on the confusion matrix provided : 14

- (a) Accuracy $\rightarrow 0.5$
- (b) Precision $\rightarrow 0.10$
- (c) Recall (Sensitivity) $\rightarrow 0.83$

(d) Specificity $\rightarrow 0.58$

(e) F1 Score $\rightarrow 0.863$

(f) Misclassification rate $\rightarrow 0.77$

(g) Macro Recall. $\rightarrow 1.25$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{x - \mu}{\sigma}$$